

Machine Learning Introduction

A practical introduction to learning from data

What does a model learn?

Machine learning is a way to build software that discovers useful patterns in examples. A model receives data, applies a mathematical rule, and returns a prediction. Training adjusts that rule so predictions fit the examples more closely. The goal is not simply to remember the training data. A useful model should generalize to new observations collected under similar conditions.

Imagine predicting the demand for a neighborhood bicycle service. Historical rentals, temperature, day of the week, and rainfall can become features. The number of rentals is the target. Before training a model, we inspect missing values, unusual measurements, and changes in how the data was collected. Clean data and a clear question often matter more than a complicated algorithm.

Learning with examples

Supervised learning uses examples with known targets. Regression predicts a quantity such as demand, while classification predicts a category such as whether an order needs review. Unsupervised learning explores data without predefined targets. Clustering, for example, can group observations that share similar characteristics. These groups are useful descriptions, but they do not automatically explain why a pattern exists.

A baseline is a simple prediction rule that gives us a starting point. For bicycle rentals, yesterday's demand or an average for the same weekday might be a useful baseline. A complex model should improve on that baseline in a meaningful way. We compare models with a metric that reflects the actual decision, rather than choosing whichever score looks most impressive.

Evaluating responsibly

We separate training data from evaluation data before fitting transformations or selecting features. Information from the evaluation set must not leak into training. When observations follow a timeline, a chronological split may be more realistic than a random split. Cross-validation can estimate how much results vary across different subsets, but its design must respect the structure of the data.

Accuracy is only one part of evaluation. We also look at error patterns, performance for relevant groups, stability, and the cost of a wrong prediction. A model that performs well on average may still fail in important situations. Documenting assumptions helps the next developer understand what was tested and what remains uncertain.

From experiment to useful system

A reliable learning system includes more than its model. Data validation, versioned experiments, clear outputs, and monitoring make results easier to maintain. We record the dataset, the training configuration, the evaluation method, and the limitations. When data changes, previous performance estimates may stop being representative.

For a first project, start with a small dataset and an understandable baseline. Build the simplest complete workflow: inspect data, split it, train, evaluate, and explain the result. The strongest project tells a clear story about a real question and shows how evidence supports its conclusions.